Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2025	ENGI3291-DE01

Title: (BEng/MEng)

Thermodynamics And Fluid Mechanics 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.* • *In line with the Department of Engineering document entitled "Student Guidance on the Use of Generative Artificial Intelligence and Related Technology for the Department of Engineering" policy in assessments, the use of generative artificial intelligence tools and related technologies is strictly prohibited for this exam.*

Revision:	

Q1

(a) The compressor of a jet engine is often designed as axial-flow type, and consists of multiple repeating stages. Consider one compressor stage having constant axial velocity.

i. Sketch a cascade view of the stage and the corresponding velocity triangles. In the cascade view, include the known velocity vectors, their angles and the numbering system. **[20%]**

ii. Write down the mathematic expressions for flow coefficient ϕ, stage loading coefficient ψ, and reaction R for this particular compressor stage, and show that

$$R = 1 - \frac{\phi^2}{2\psi}\left(\tan^2\alpha_2 - \tan^2\alpha_3\right),$$

where α_2 and α_3 are the angle of the absolute velocity with respect to the axial direction at the rotor exit and the stator exit, respectively. **[30%]**

iii. Hence show that the above relation can further be simplified to

$$R = 1 - \frac{\phi}{2}\left(\tan\alpha_1 + \tan\alpha_2\right),$$

where α_1 is the angle of the absolute velocity with respect to the axial direction at the stage inlet. **[20%]**

(b) A low pressure compressor within a jet engine consists of nine repeating stages with constant axial velocity. It operates with a mass flow of 30 kg s^{-1} and uses 7.5 MW of mechanical power. The rotor in each compressor stage turns the absolute flow velocity from 13° at rotor inlet to 30° at rotor outlet. The rotor mean radius is 0.5 m and the rotational shaft speed is 5000 rpm. Calculate ψ, ϕ and R of each stage. Briefly comment on the values. **[30%]**

Q2

(a) Sketch a graph of the variation in binding energy per nucleon across different elements. Using this graph, name and explain the two main types of energy-releasing nuclear reactions. [**10%**]

(b) Using the equivalence between mass and energy on an atomic scale 1 u = 931.5 MeV and the data in Table Q2 calculate the Q-value for the alpha decay of U-238.

Element	Symbol	Atomic number (Z)	Atomic mass(u)
Proton	p	1	1.007277
Neutron	n	0	1.008665
Electron	e	0	0.000550
Hydrogen	^{2}H	1	2.014102
Hydrogen	^{3}H	1	3.016049
Helium	^{3}He	2	3.016029
Helium	^{4}He	2	4.002602
Radium	^{233}Ra	88	233.048065
Radium	^{234}Ra	88	234.050705
Thorium	^{234}Th	90	234.043599
Thorium	^{235}Th	90	235.047255
Thorium	^{236}Th	90	236.049657
Uranium	^{235}U	92	235.043930
Uranium	^{238}U	92	238.050788
Uranium	^{239}U	92	239.054294

Table Q2

[**15%**]

(c) A thermal reactor operates using natural uranium, consisting of 0.72% U-235 and 99.28% U-238. The average number of neutrons released per fission of U-235 is 2.5. The fast fission factor is 1.05, the resonance escape probability is 0.9, and the thermal utilisation factor is 0.8. The thermal neutron fission cross-section for U-235 is 582 barns, while the capture cross-section for U-235 and U-238 are 99 barns and 2.75 barns, respectively. Assuming neutron leakage can be neglected, calculate the reproduction constant and comment on whether a neutron chain reaction can be sustained. [**15%**]

(d) Calculate the power of the reactor described in part (c), assuming it contains 1.8×10^{25} atoms of U-235. The average energy released per fission is 200 MeV, where the unit of energy on atomic scale is 1 eV $= 1.6 \times 10^{-19}$ J. The thermal neutron flux in the reactor is 1×10^{14} neutrons/(cm^{2}s). [**15%**]

(e) Describe the two main methods of plasma confinement used to achieve nuclear fusion on Earth and explain the challenges associated with each method. [**10%**]

(f) Describe the main differences between the steam cycles of typical Pressurised Water Reactor (PWR) and Advanced Gas Cooled Reactor (AGR) nuclear power plants. [**20%**]

(g) What roles are the following types of large electric power plants likely to play in the UK's low/zero carbon future thermal power plans?

i Combined Cycle Power Plants

ii Coal Fired Power Plants

iii Nuclear Power Plants

[**15%**]

Q3

Consider a two-dimensional duct $ABCD$ shown in Fig. Q3, with a straight horizontal lower wall AD, and a straight upper wall BC but inclined upward through a 6° angle. The height of the duct entrance AB is 0.3 m. A uniform horizontal flow at $M = 2$ entres the duct from the ambient region ①, where the stagnation pressure is p_{01} and the stagnation temperature is T_{01}. The flow goes through a Prandtl-Mayer expansion fan centred at the top corner B of the entrance. The wave propagates to the bottom wall, where the first Mach wave intersects the bottom wall at point A'. Imagine a line drawn perpendicular to the lower wall at point A', and intersecting the upper wall at point B'.

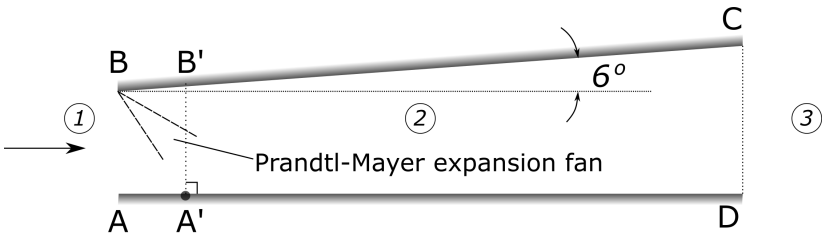


Fig. Q3: not to scale

(a) Calculate d, the length of $A'B'$, and the average flow Mach number over $A'B'$, assuming that M varies linearly along $A'B'$. Determine the static pressure at point B'. **[35%]**

(b) The portion of the duct $A'B'CD$, which is marked as ②, is characterised by a two-dimensional duct with a roughness factor $f = 0.003$. The length of this portion $A'D = 2$ m. If wall $B'C$ can be approximated to be parallel with AD, determine the Mach number and the stagnation pressure in terms of p_{01} at the end of region ②. Based on the pressure at corner C, determine the pressure force per unit span (into the screen) on the wall BC in terms of p_{01}, assuming pressure varies linearly along this wall. **[40%]**

(c) Downstream of CD is region ③, where the flow direction returns to purely horizontal everywhere. Characterise the flow feature at the end of region ②. Calculate the average Mach number in region ③, assuming it is the average value between the top (corner C) and the bottom (corner D) of the region. Hence, determine the stagnation temperature in terms of T_{01}, and the stagnation pressure in terms of p_{01}, in region ③. **[25%]**

END